

ASSIGNMENT-3

1. Write a program in C to extract a substring from a given string.

```
#include <stdio.h>

#include <string.h>

int extract(int from, int to, char *string, char *subString)

{

    int i=0, j=0;

    //get the length of the string.

    int length = strlen(string);

    if( from > length || from < 0 ){

        printf("The index 'from' is invalid\n");

        return 1;

    }

    if( to > length ){

        printf("The index 'to' is invalid\n");

        return 1;

    }

    for( i = from, j = 0; i <= to; i++, j++){

        subString[j] = string[i];

    }

    return 0;
```

```
int main()

{

    char string[100];

    char subString[50];

    int from,to;


    printf("Enter a string: ");

    fgets(string,100,stdin);

    printf("Enter the index 'from': ");

    scanf("%d",&from);

    printf("Enter the index 'to': ");

    scanf("%d",&to);


    printf("The string is : %s",string);


    if( extract(from, to, string, subString) == 0 )

        printf("The sub-string is : %s", subString);

    else

        printf("ERROR!\n");


    return 0;

}
```

OUTPUT:

Enter a string: Welcome to StackHowTo

Enter the index 'from': 11

Enter the index 'to': 20

The string is : Welcome to StackHowTo

The sub-string is : StackHowTo

2. Write a C program to check whether a given substring is present in the given string

```
#include <stdio.h>
```

```
void main()
```

```
{
```

```
    char str[80],search[20];
```

```
    int c1=0,c2=0,i,j,flg;
```

```
    printf("\n\nCheck whether a given substring is present in the given string\n");
```

```
    printf("-----\n");
```

```
    printf("Input the string : ");
```

```
    fgets(str, sizeof str, stdin);
```

```
    printf("Input the substring to be search : ");
```

```
    fgets(search, sizeof search, stdin);
```

```
    while (str[c1]!='\0')
```

```
        c1++;
```

```

        c1--;

while (search[c2]!='\0')
    c2++;
    c2--;

for(i=0;i<=c1-c2;i++)
{
    for(j=i;j<i+c2;j++)
    {
        flg=1;
        if (str[j]!=search[j-i])
        {
            flg=0;
            break;
        }
    }
    if (flg==1)
        break;
}
if (flg==1)
    printf("The substring exists in the string.\n\n");
else
    printf("The substring is not exists in the string. \n\n");
}

```

OUTPUT:

```
Input the string : This is a test string.
Input the substring to be search : search
The substring is not exists in the string.
```

3. Write a program in C to read a sentence and replace lowercase characters by uppercase and vice-versa

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void main()
{
    char str[100];
    int ctr, ch, i;

    printf("\n\nReplace lowercase characters by uppercase and vice-versa :\n");
    printf("-----\n");

    printf("Input the string : ");
    fgets(str, sizeof str, stdin);

    i = strlen(str);

    ctr = i; /*shows the number of chars accepted in a sentence*/

    printf("\nThe given sentence is  : %s", str);

    printf("After Case changed the string is: ");
    for(i=0; i < ctr; i++)
    {
        ch = islower(str[i]) ? toupper(str[i]) : tolower(str[i]);
        putchar(ch);
    }
    printf("\n\n");
}
```

OUTPUT:

```
Input the string : This Is A Test String

The given sentence is  : This Is A Test String
After Case changed the string is: tHIS iS a tEST sTRING
```

4. Write a program in C to find the number of times a given word 'the' appears in the given string.

```
#include <stdio.h>
```

```

#include <string.h>
void main()
{
    int ctr=0,i,freq=0;
    int t,h,e,spc;
    char str[100];

    printf("\n\nFind the number of times the word 'the ' in any combination appears :\n");
    printf("-----\n");

    printf("Input the string : ");
    fgets(str,sizeof str,stdin);

    ctr=strlen(str);

    /*Counts the frequency of the word 'the' with a trailing space*/
    for(i=0;i<=ctr-3;i++)
    {
        t=(str[i]=='t' | | str[i]=='T');
        h=(str[i+1]=='h' | | str[i+1]=='H');
        e=(str[i+2]=='e' | | str[i+2]=='E');
        spc=(str[i+3]==' ' | | str[i+3]=='\0');
        if ((t&&h&&e&&spc)==1)
            freq++;
    }
    printf("The frequency of the word \'the\' is : %d\n\n",freq);
}

```

OUTPUT:

```

Input the string : The stering where the word the present more then onces.
The frequency of the word 'the' is : 3

```

5. Write a program in C to Find the Frequency of Characters

```
#include <stdio.h>
```

```
int main() {
```

```
    char str[1000], ch;
```

```
    int count = 0;
```

```
    printf("Enter a string: ");
```

```
    fgets(str, sizeof(str), stdin);
```

```
    printf("Enter a character to find its frequency: ");
```

```
    scanf("%c", &ch);
```

```
    for (int i = 0; str[i] != '\0'; ++i) {
```

```
        if (ch == str[i])
```

```
            ++count;
```

```
    }
```

```
    printf("Frequency of %c = %d", ch, count);
```

```
    return 0;
```

```
}
```

Output

Enter a string: This website is awesome.

Enter a character to find its frequency: e

Frequency of e = 4